GERMÁN KÜBER

Senior Backend Engineer — Production Python — Real-Time AI & Voice Pipelines

Las Palmas, Spain · EU Citizen — authorized to work across the EU/EEA, no visa sponsorship required
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PROFESSIONAL SUMMARY

Senior backend engineer with 15+ years building production systems and a deep focus over the last several years on production Python (FastAPI, Django, Flask) for real-time AI products. Shipped a real-time audio AI pipeline at KliverAI on OpenAI Realtime API with sub-200ms streaming latency and a WebSocket-based transcription pipeline capturing live meeting audio — directly applicable to ambient AI / scribe products. Operated production Python and Rust services on Docker and Kubernetes / AWS at high scale (Borderless Capital: $50M+ daily transactional flow). Comfortable across adjacent stacks including **TypeScript / Node.js** and .NET. Customer-facing impact orientation; comfortable with ambiguity and fast iteration; very much a fit for ship-it-now environments. EU citizen (Spanish nationality) with full authorization to work across the EU/EEA — no visa or sponsorship required — open to self-funded relocation anywhere in the EU.

CORE EXPERTISE

Production Python: FastAPI, Flask, Django, asyncio, async / await, Cython, PyO3 bindings, Pydantic, SQLAlchemy, REST and WebSocket APIs, background workers, structured logging, testing (pytest)

Real-Time AI & Voice Pipelines: OpenAI Realtime API, WebSocket streaming, SignalR, sub-200ms audio pipelines, real-time transcription capture, real-time evaluation pipelines, LLM orchestration (OpenAI, Anthropic, Gemini), multi-provider routing, prompt engineering, RAG, multi-agent (LangGraph), MCP server / client

Cloud & Infrastructure: AWS, Docker, Kubernetes (manifests, deployments, services), CI / CD pipelines, Linux, Infrastructure as Code (Terraform), observability stacks

Data & Storage: PostgreSQL (OLTP, jsonb, semantic cache backing store), caching and pub/sub (Redis-style), high-throughput data pipelines

Backend Engineering: Event-driven microservices, async coordination, fault tolerance, robust APIs and abstractions, Domain-Driven Design, customer-facing systems at scale

Agent Engineering: Production AI agents, agentic workflows, LangChain, LangGraph, Microsoft Agent Framework, A2A protocol, eval frameworks (LangSmith, Langfuse, RAGAS, DeepEval), context engineering, knowledge graphs (Cognee, Graphiti), LoRA fine-tuning, semantic caching, human-in-the-loop

Paradigms, Design & Practices: OOP, Functional Programming, Design Patterns, Architecture Patterns, SOLID, data structures, algorithm design, clean code, Test-Driven Development (TDD), pair programming, code review, mentoring, ADRs, CI / CD, Infrastructure as Code (Terraform)

Observability & Reliability: Prometheus, OpenTelemetry, distributed tracing, structured logging, 50+ production metrics, 24 / 7 on-call rotation, post-mortems, incident response

Adjacent Stacks: TypeScript / Node.js, Rust (tokio, async), .NET / C#, Go (reverse-engineering Geth internals)

PROFESSIONAL EXPERIENCE

Independent AI Consultant — Agent Engineering & Enterprise AI Delivery

Self-employed (autónomo, Spain) — Mar 2026 – Present — Remote, client under NDA

- Forward-deployed AI consultant embedded with an enterprise engineering organization: run discovery with business and technical stakeholders, architect the solution, and own delivery from pilot through production and continuous iteration.
- Design and ship production agentic workflows in Python on LangChain, LangGraph and Microsoft Agent Framework, testing each graph, running eval suites against it, and monitoring it live in production.
- Own the RAG platform end-to-end for internal company documentation — retrieval optimization, chunking, context engineering, and regression test suites gating every pipeline change.
- Fine-tuned models with LoRA on internal company documents deployed through Azure AI Foundry; cut cost and latency through semantic caching, semantic memory, and dynamic model routing.

Lead AI Architect & Co-Founder

KliverAI (Own Venture, closed Feb 2026) — Mar 2025 – Feb 2026 (full-time) — Remote, Spain

- Designed and shipped a real-time audio AI pipeline on OpenAI Realtime API with SignalR streaming, achieving sub-200ms end-to-end latency across concurrent user sessions — capturing live conversation, processing it through LLMs, and returning structured output in real time.
- Rewrote the real-time layer from app-embedded SignalR to Azure SignalR Service with Redis-backed state, scaling concurrent users by 500% while holding sub-200ms latency.
- Benchmarked TTS / STT services (ElevenLabs, Deepgram, AssemblyAI, OpenAI Whisper, Play.ht) on latency, cost, and quality — getting latency to 90ms and word accuracy to nearly 95%.
- Built a real-time transcription capture pipeline: a Chrome Extension streams live Google Meet audio over WebSocket into a Python backend, which then feeds an automated AI evaluation pipeline.
- Built backend services in Python (FastAPI, asyncio) for AI integrations and data pipelines, alongside Node.js / TypeScript services for the API and streaming layer; deployed everything on Docker and Kubernetes on AWS.
- Integrated three LLM providers (OpenAI, Anthropic, Gemini) with dynamic model routing, picking the best model per task to cut inference spend.
- Built an adaptive multi-agent platform on LangGraph with multi-step branching logic and RAG-powered personalization.
- Implemented MCP (Model Context Protocol) servers and clients to standardize tool integration across services.
- Established the observability stack with OpenTelemetry, distributed tracing, structured logging, and LLM-specific tracing (Langfuse, LangSmith) plus prompt evaluation (Braintrust, PromptLayer, Promptfoo).
- Hired and led a distributed team of up to 3 engineers, owning technical interviews, hiring, and architectural reviews — hands-on in the codebase throughout.

Fractional CTO (Part-time, hands-on)

Rocking Product — Apr 2025 – Dec 2025 (part-time, parallel to KliverAI) — Remote

- Came in as Fractional CTO to an 8-engineer team and restructured it in the first quarter without stepping away from the code; ran technical interviews and hiring decisions.

- Built an image embeddings service (Azure AI Vision and CLIP) to improve product matching across brand comparisons, raising accuracy and lifting the client's revenue by 40%.
- Cut production bugs by 70% through team training and automated processes (testing, CI / CD, code review).
- Shipped customer-facing AI features in production using Python (FastAPI, LangChain) and Node.js / TypeScript, establishing the team's first LLM evaluation and monitoring practices.
- Optimized LLM API usage and cloud infrastructure (AWS, Docker, Kubernetes) through model selection, prompt compression, and semantic caching — cutting combined LLM API and cloud spend by 45%.
- Designed a reusable base architecture on LangChain and LangGraph with Langfuse / LangSmith monitoring; introduced Spec-Driven Development (SDD); proposed and led a full-stack migration from a fragmented JavaScript codebase to .NET.

Lead Backend Engineer (Distributed Systems, Python & Rust)
Borderless Capital / CTF Capital — Sep 2021 – Mar 2025 — Remote

- Operated production backend services processing 10K+ opportunities per day from $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L with execution under 10ms p99.
- Migrated the system from Python / Cython to Rust and took it to 20x more transactions per second.
- Built Python services (endpoints, business logic, external integrations) running on Docker and Kubernetes in production, alongside the Rust core; owned K8s manifests and deployment configuration for the services I shipped.
- Built Python / Cython and PyO3 FFI bindings exposing Rust performance-critical code to higher-level Python orchestration layers, eliminating GC pauses across language boundaries.
- Used Postgres as the system of record for pipeline state, opportunities, and execution outcomes, and Redis-style caching both for low-latency caching and as a pub / sub backbone between services.
- Engineered low-latency observability with Prometheus and custom telemetry, tracking 50+ real-time metrics across the production pipeline.
- Ran technical due diligence on 5+ projects that ended up getting funded — architecture, risk, and feasibility — acting as the CTO's technical reference on the low-level side.
- Owned 24 / 7 production reliability and incident response; acted as technical lead across the blockchain team, which grew from 4 to 10 engineers — code review, mentoring, and hiring.

Senior Software Architect (Backend, Event-Driven)
Paramo Technology — Oct 2020 – Sep 2021 — Remote

- Owned the end-to-end architecture of a high-traffic online casino (iGaming) platform handling real-time money operations and regulatory reporting, working directly with the CTO.
- Spotted the issues slowing teams down and designed action plans that raised productivity by 40%, measured by features shipped and fewer errors.
- Designed event-driven microservices in .NET, implemented distributed tracing (OpenTelemetry, Application Insights, Datadog), brought in Kubernetes to scale, and built automated deploy pipelines.
- Led part of the hiring and designed the technical entry exam; delivered internal training on architecture, design patterns, and operational practices.

Software Architect
CloudX (Software Consultancy) — Sep 2017 – Oct 2020 — Remote / Buenos Aires (first employee)

- Was the company's first employee; led architecture decisions across several enterprise product lines, designing microservices backends with Domain-Driven Design.
- Built a distributed system over geospatial data with PostGIS for a UK client; delivered technical training and conference talks on backend architecture.

Software Architect / Team Lead — Earlier Roles
Majestic Hotel Group (2014–2017) · Plus54 (2011–2014) · Acsys (2009–2011) · IBM (2008–2009) — Argentina

- Full-stack features for internal hotel management platforms, first as Team Lead then architect; led migration from AngularJS to Angular 2 and redesigned the data model toward event-oriented architecture. Earlier: .NET web applications, internal tooling and mobile components end to end.

KEY ACHIEVEMENTS

Real-time audio AI pipeline at KliverAI — OpenAI Realtime API + SignalR + WebSocket transcription capture, sub-200ms streaming latency across concurrent sessions.
Production backend at Borderless — Python and Rust services on Docker / Kubernetes / AWS, processing $50M+ in daily transactional flow and contributing $100K+ in attributable annual P&L at sub-10ms p99.
Production AI agents (Independent Consulting) — Agentic workflows in Python on LangChain, LangGraph and Microsoft Agent Framework, eval-gated and monitored live in production for an enterprise client.
Microsoft MVP, Developer Technologies — 8 consecutive years (2017–present). **80+ International Technical Talks** — conferences and meetups across LATAM and Europe. **Net-Baires Founder** — Argentina's largest .NET developer community, 100K+ members across 10+ years.

EDUCATION & CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) · Azure Associate (AZ-104 / AZ-204 level) — Microsoft
Microsoft MVP, Developer Technologies — Feb 2017 – Present (8 consecutive years)
Polkadot Builder Program — Polkadot Blockchain Academy, Lisbon, Portugal (2026)
Systems Engineer — Universidad Abierta Interamericana (UAI), Buenos Aires, Argentina (2017–2021)

OPEN SOURCE, WRITING & COMMUNITY

YouTube channel (youtube.com/@germankuber) — technical deep-dives on AI, Python, Rust, and software architecture.
Net-Baires (.NET Community) — Founder (2015–present), 100K+ members. **Crypto Para Mortales** — Founder (2023–present), Spanish-language blockchain education conference. **Patrones Para Mortales** — Founder (2020–present), educational content on software design.

SPOKEN LANGUAGES

Spanish: Native — English: Full Professional Proficiency — daily use in international remote teams